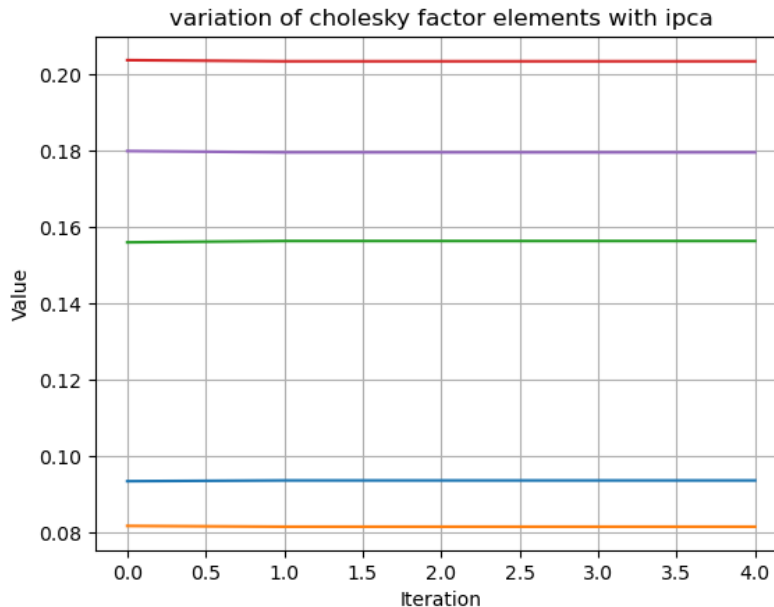


CH5440 ASSIGNMENT 5 : IPCA

[ME20B072 GIRISH MADHAVAN]

- a. Assuming the errors in different variables are independent, apply IPCA to estimate diagonal error variances and identify the linear steady state model relating the flow variables (assuming that the number of constraints is known). For this purpose use the function `stddest.m`, which uses a least squares procedure for estimating diagonal error variances. This function takes the constraint model and data as inputs and returns the standard deviation of errors as output.

SOLN :



- The above convergence plot is obtained for **flowdata1.mat** data

CHOLESKY MATRIX :

```
[[0.0937 0. 0. 0. 0. ]
 [0. 0.0816 0. 0. 0. ]
 [0. 0. 0.1564 0. 0. ]
 [0. 0. 0. 0.2034 0. ]
 [0. 0. 0. 0. 0.1796]]
```

CONVERGED CONSTRAINT MATRIX :

```
[[ -3.4973  0.003  2.7449 -2.7619  3.4889]
 [ 4.6006  5.8715 -3.712  -2.1815  1.2529]
 [ -3.4973  0.0119 -2.7449  2.7619  3.4889]]
```

CONVERGED EIGENVECTORS :

```
[[ -37.3223  0.0365 17.5516 -13.5756 19.4262]
 [ 49.0967 71.9539 -23.7353 -10.7229  6.9761]
 [ -37.3224  0.146 -17.5517 13.5756 19.4262]]
```

TRUE REGRESSION MATRIX :

```
[[1. 1.]
 [1. 1.]
 [1. 0.]]
```

CONVERGED REGRESSION MATRIX :

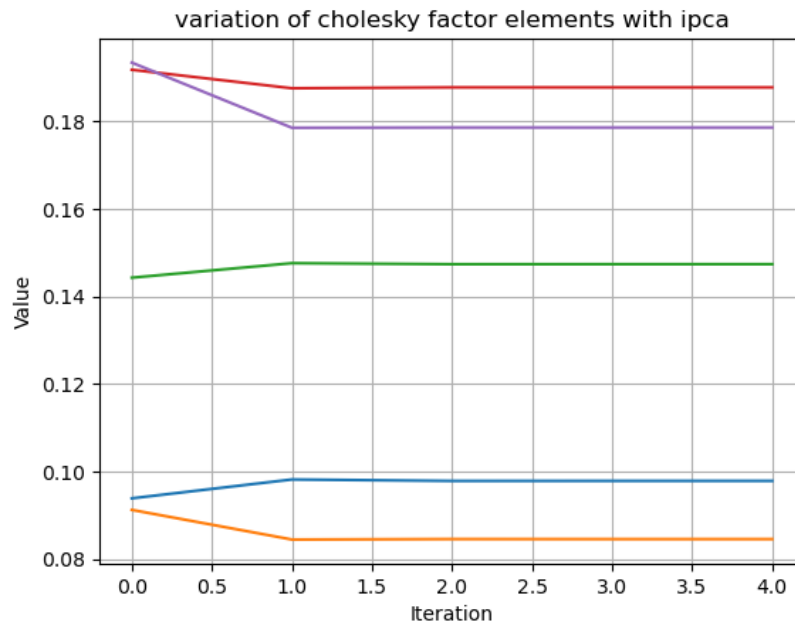
```
[[ 0.996  0.9987]
 [ 0.9899  0.9909]
 [ 1.0024 -0.0021]]
```

ABSOLUTE DIFFERENCE BETWEEN THE REGRESSION MATRICES :

```
[[0.004 0.0013]
 [0.0101 0.0091]
 [0.0024 0.0021]]
```

- From the above converged values we can observe that the error in the regression matrix is minimal
- This is because the error variances in the data are not correlated and hence there is no non diagonal elements in the error covariance matrix
- The converged cholesky matrix is very close to the standard deviation matrix

[[0.1 0.08 0.15 0.2 0.18]]



- The same set of calculations have been performed for flowdata2.mat and the results are shown

CHOLESKY MATRIX :

```
[[0.0978 0.      0.      0.      0.      ]
 [0.      0.0845 0.      0.      0.      ]
 [0.      0.      0.1474 0.      0.      ]
 [0.      0.      0.      0.1878 0.      ]
 [0.      0.      0.      0.      0.1786]]
```

CONVERGED CONSTRAINT MATRIX :

```
[[ 3.5064  0.3628 -3.0813  2.8829 -3.4618]
 [-4.5483 -6.0744  3.4976  2.4473 -1.3912]
 [ 3.5056 -0.3143  3.0812 -2.883  -3.463  ]]
```

CONVERGED EIGENVECTORS :

```
[[ 35.8466  4.2932 -20.9105  15.3536 -19.3835]
 [-46.4988 -71.8769  23.7359  13.0338  -7.7897]
 [ 35.839  -3.7195  20.9097 -15.3545 -19.3903]]
```

TRUE REGRESSION MATRIX :

```
[[1. 1.]
 [1. 1.]
 [1. 0.]]
```

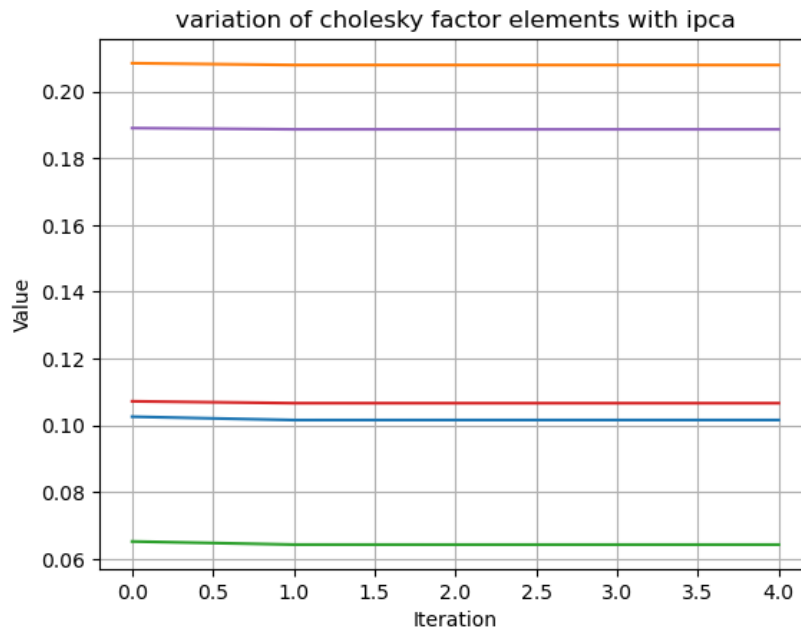
CONVERGED REGRESSION MATRIX :

```
[[0.9746 1.0422]
 [1.0413 0.9965]
 [1.0126 0.007  ]]
```

ABSOLUTE DIFFERENCE BETWEEN THE REGRESSION MATRICES :

```
[[0.0254 0.0422]
 [0.0413 0.0035]
 [0.0126 0.007  ]]
```

- From the error in regression matrix we can see that as compared to the previous case the error is now 10 times as that of the previous value, this might be due to some problems with our assumption that the covariance matrix is diagonal



- Finally for the flowdata3.mat, although the values have converged the error is higher than first case suggesting the presence of correlation in the errors

CHOLESKY MATRIX :

```
[[0.1016 0.      0.      0.      0.      ]
 [0.      0.208  0.      0.      0.      ]
 [0.      0.      0.0642 0.      0.      ]
 [0.      0.      0.      0.1066 0.      ]
 [0.      0.      0.      0.      0.1887]]
```

CONVERGED CONSTRAINT MATRIX :

```
[[ 1.3635 -2.7589 -0.1856  2.8615 -3.9604]
 [-5.3352 -2.9834  0.184   2.6513  2.6033]
 [-1.0692 -1.5171  8.6109 -7.1017 -0.5082]]
```

CONVERGED EIGENVECTORS :

```
[[ 13.4232 -13.2661 -2.8885  26.8367 -20.99  ]
 [-52.5225 -14.3456  2.8642  24.8654  13.7976]
 [-10.5255 -7.2951 134.0253 -66.6035 -2.6932]]
```

TRUE REGRESSION MATRIX :

```
[[1. 1.]
 [1. 1.]
 [1. 0.]]
```

CONVERGED REGRESSION MATRIX :

```
[[0.9796 1.0379]
 [0.966  1.0442]
 [0.9963 0.0092]]
```

ABSOLUTE DIFFERENCE BETWEEN THE REGRESSION MATRICES :

```
[[0.0204 0.0379]
 [0.034  0.0442]
 [0.0037 0.0092]]
```

- b. Assume that the number of constraints is unknown and estimate it using hypothesis test for equality of eigenvalues, along with the error standard deviations and model constraints.

SOLN : In this Question we run IPCA assuming the number of constraints to be some number between 1 and 4. After this we perform hypothesis testing to verify that we have correctly determined the number of constraints.

FLOW DATA USED	OBTAINED NUMBER OF CONSTRAINTS AFTER IPCA
flowdata1.mat	3
flowdata2.mat	3
flowdata1.mat	1

flowdata1.mat : we can observe at $m = 3$ the correct number of constraints are detected

Number of Constraints (m) for ipca : 1

4 | Test Statistic : 12895.038 Chi2 Value : 16.919

3 | Test Statistic : 2362.913 Chi2 Value : 11.07

2 | Test Statistic : 1065.342 Chi2 Value : 5.991

Number of Constraints (m) for ipca : 2

4 | Test Statistic : 11990.364 Chi2 Value : 16.919

3 | Test Statistic : 1622.971 Chi2 Value : 11.07

2 | Test Statistic : 1231.192 Chi2 Value : 5.991

Number of Constraints (m) for ipca : 3

4 | Test Statistic : 10171.127 Chi2 Value : 16.919

Detected Number of Constraints : 3

Number of Constraints (m) for ipca : 4

4 | Test Statistic : 6370.76 Chi2 Value : 16.919

3 | Test Statistic : 4330.562 Chi2 Value : 11.07

2 | Test Statistic : 101.96 Chi2 Value : 5.991

flowdata2.mat : Again for $m = 3$ the correct number of constraints are detected

Number of Constraints (m) for ipca : 1

4 Test Statistic : 5672.55	Chi2 Value : 16.919
------------------------------	---------------------

3 Test Statistic : 4068.742	Chi2 Value : 11.07
-------------------------------	--------------------

2 Test Statistic : 1816.423	Chi2 Value : 5.991
-------------------------------	--------------------

Number of Constraints (m) for ipca : 2

4 Test Statistic : 1502.929	Chi2 Value : 16.919
-------------------------------	---------------------

3 Test Statistic : 913.843	Chi2 Value : 11.07
------------------------------	--------------------

2 Test Statistic : 124.262	Chi2 Value : 5.991
------------------------------	--------------------

Number of Constraints (m) for ipca : 3

4 Test Statistic : 1596.41	Chi2 Value : 16.919
------------------------------	---------------------

Detected Number of Constraints : 3

Number of Constraints (m) for ipca : 4

4 Test Statistic : 829.033	Chi2 Value : 16.919
------------------------------	---------------------

3 Test Statistic : 277.341	Chi2 Value : 11.07
------------------------------	--------------------

2 Test Statistic : 29.97	Chi2 Value : 5.991
----------------------------	--------------------

- The regression matrix and converged matrices remain the same for flowdata1 and flowdata2 after performing hypothesis testing because the number of constraints are identified correctly

flowdata3.mat : In this case we can observe that the correct number of constraints are not being detected, instead it fails hypothesis testing every time and tells that there is only one constraint between the variables while checking for the equality of eigenvalues

Number of Constraints (m) for ipca : 1

4 Test Statistic : 4670.254	Chi2 Value : 16.919
-------------------------------	---------------------

3 Test Statistic : 2629.441	Chi2 Value : 11.07
-------------------------------	--------------------

2 Test Statistic : 33.273	Chi2 Value : 5.991
-----------------------------	--------------------

Number of Constraints (m) for ipca : 2

4 Test Statistic : 9573.539	Chi2 Value : 16.919
-------------------------------	---------------------

3 Test Statistic : 2951.348	Chi2 Value : 11.07
-------------------------------	--------------------

2 Test Statistic : 1121.856	Chi2 Value : 5.991
-------------------------------	--------------------

Number of Constraints (m) for ipca : 3

4 Test Statistic : 6336.625	Chi2 Value : 16.919
-------------------------------	---------------------

3 Test Statistic : 263.25	Chi2 Value : 11.07
-----------------------------	--------------------

2 Test Statistic : 213.558	Chi2 Value : 5.991
------------------------------	--------------------

Number of Constraints (m) for ipca : 4

4 Test Statistic : 3754.711	Chi2 Value : 16.919
-------------------------------	---------------------

3 Test Statistic : 837.756	Chi2 Value : 11.07
------------------------------	--------------------

2 Test Statistic : 76.987	Chi2 Value : 5.991
-----------------------------	--------------------

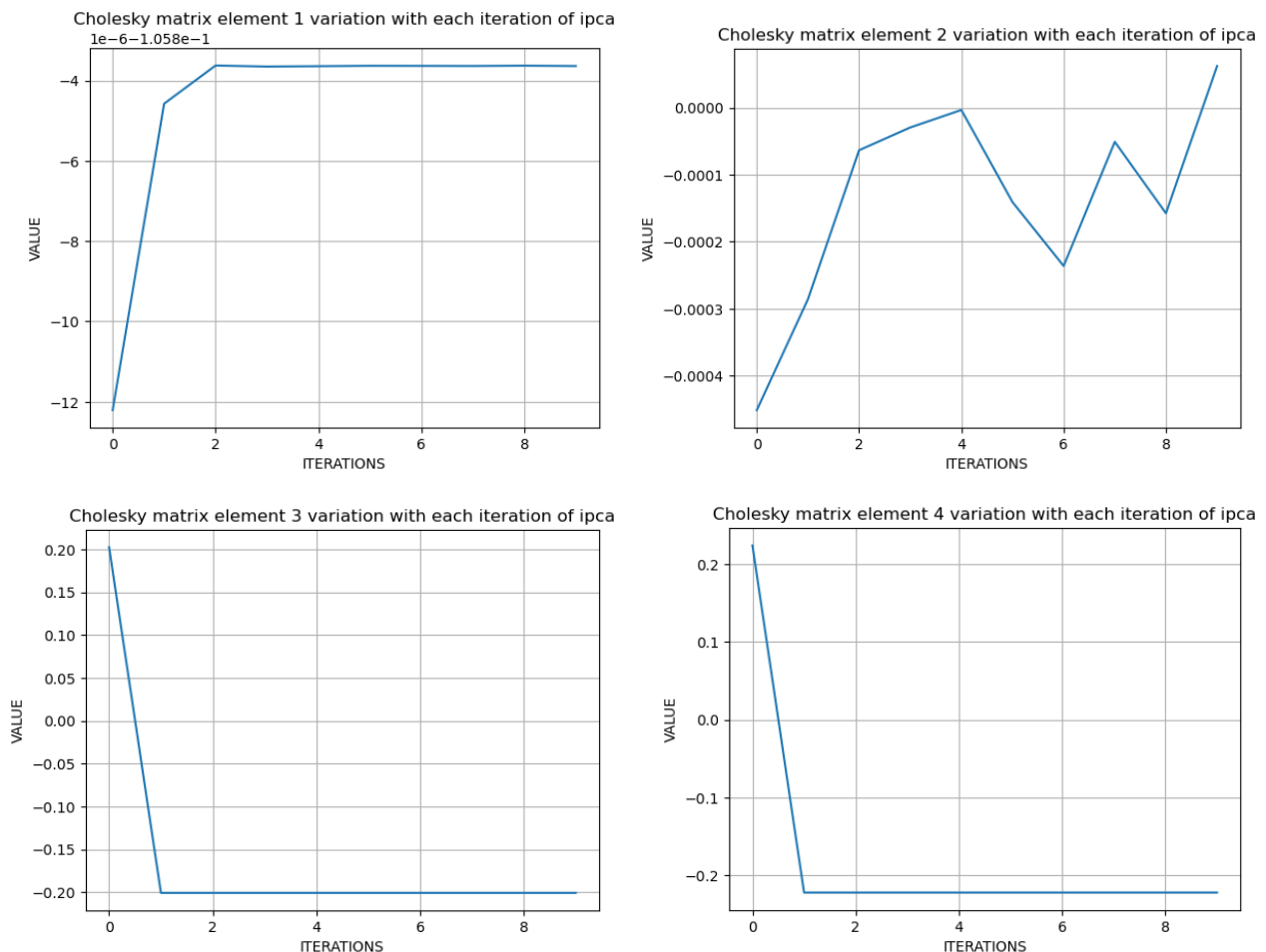
- c. Assume that the errors in measurement of variables F3 and F4 are correlated. In this case, the error covariance matrix is not diagonal, but also contains an off-diagonal element corresponding to the element (3, 4) and (4,3) in the error covariance matrix. Modify the `stdest.m` function to estimate the diagonal and off-diagonal elements of the error covariance matrix and use it to extend the IPCA method to simultaneously estimate the number of constraints, elements of the error covariance matrix and constraint model

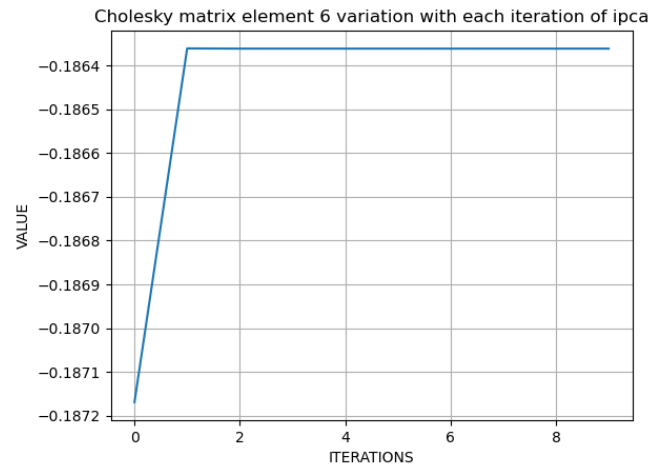
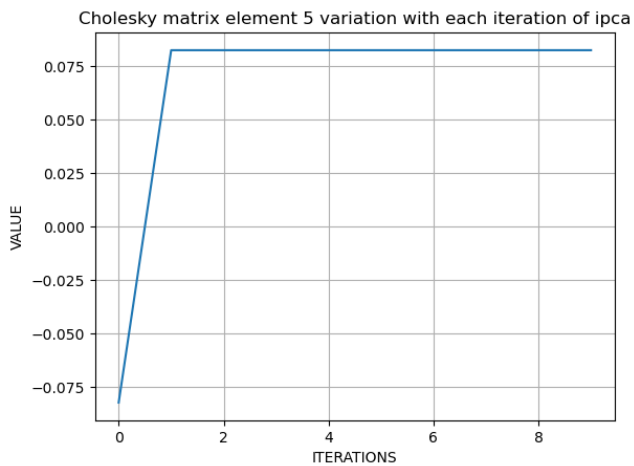
SOLN :

PROCEDURE :

- Assume that there are three constraints in the constraint matrix and the cholesky matrix (**L**) to be a diagonal matrix of 1 and dimensions $p \times p$ (p is the number of variables)
- Compute Constraint matrix (**As**) from the Mean shifted data and scaled matrix (**Zs**) using PCA
- Note that for the first iteration the **Zs** is same as **Zm** (mean shifted values)
- Use **As** along with **Zs** to compute the residuals matrix (**R**)
- Compute the constraint matrix (**A**) in the unscaled domain by post multiplying **As** with inverse of **L**
- Now pass **A** and **R** in the optimization function that will minimise the maximum likelihood of the residuals with respect to the new estimate of **L** (output)
- Now use this new **L** to scale the data matrix and perform PCA to get new **As** and so on
- Repeat the above steps until the errors fall below a particular tolerance

CONVERGENCE PLOTS FOR THE ELEMENTS OF THE CHOLESKY MATRIX [L] :





CHOLESKY MATRIX :

```

[[-0.10580364  0.          0.          0.          0.          ]
 [ 0.          0.00006253  0.          0.          0.          ]
 [ 0.          0.         -0.20082279  0.          0.          ]
 [ 0.          0.         -0.22195259  0.08228568  0.          ]
 [ 0.          0.          0.          0.         -0.18636197]]

```

STANDARD DEVIATION VALUES FOR flowdatat3 :

```
[[0.1  0.08 0.15 0.2  0.18 0.12]]
```

CONVERGED CONSTRAINT MATRIX :

```

[[-3.29771139  0.1231134  8.09813579 -8.23614353  3.30018996]
 [ 3.50590075  4.63395305 -7.78293632  3.13618942  1.12918676]
 [ 3.29769289  0.18144123  8.09809006 -8.23612332 -3.30019748]]

```

CONVERGED EIGENVECTORS :

```

[[ 0.34890984 -0.00001938  0.20174314 -0.67771667 -0.61502987]
 [-0.37093703 -0.00072955  0.86690513  0.25806348 -0.21043746]
 [-0.34890788 -0.00002857  0.20174784 -0.67771501  0.61503127]]

```

TRUE REGRESSION MATRIX :

```

[[1. 1.]
 [1. 1.]
 [1. 0.]]

```

CONVERGED REGRESSION MATRIX :

```

[[0.98615172 1.00055523]
 [0.96962429 1.00227693]
 [0.99924115 0.00883317]]

```

ABSOLUTE DIFFERENCE BETWEEN REGRESSION MATRICES :

```

[[0.01384828 0.00055523]
 [0.03037571 0.00227693]
 [0.00075885 0.00883317]]

```


HYPOTHESIS TESTING :

4 Test Statistic : 6085.81	Chi2 Value : 16.919
3 Test Statistic : 0.824	Chi2 Value : 11.07

Number of Constraints : 3

CONCLUSION :

- One reason for failing to converge to the proper standard deviation values could be because the actual cholesky elements are close to each other in values and don't show much variation
- But still the errors obtained in the regression matrix is much lower than the ones obtained in the first question while considering no correlation between the errors for flowdata3.mat, This is because in TLS Objective function only the ratio of the standard deviations are important and the values themselves are not needed. So in our case it might be possible that these ratios were close enough but the values are not.
- Its because of this reason we use the regression matrix to compare the performance of each IPCA algorithm on various cases
- This entire process was computationally intensive as it requires minimization of likelihood function unlike the previous cases where we just performed matrix operations to arrive at the solution
- The time taken for this Computation : About 8 minutes for 10 iterations